

TASK 7: TIME SERIES ANALYSIS

Dataset: Stock Price Dataset

Objective:

Analyze time-based trends

Tasks:

- Trend analysis
- Seasonality detection
- (Optional) Forecast future values

Output:

Time-based insights

Tesla Stock Price Time Series Analysis Report

1. Executive Summary

This report details a comprehensive time series analysis of Tesla stock data spanning from September 2019 to April 2022. The analysis aimed to understand historical price trends, detect underlying seasonality, model volatility, and generate a short-term price forecast. Key findings include a historic bull run from March 2020 to November 2021, resulting in a staggering **1,925.84% total return**, followed by a significant market correction. The ARIMA forecasting model suggests a period of continued consolidation with a slight downward pressure in the immediate future.

2. Data Overview & Preparation

The dataset, TSLA.csv, contains 639 trading days of historical data with the following features: Date, High, Low, Open, Close, Volume, and Adj Close.

- **Preprocessing Steps:**

1. The Date column was converted to a datetime format.
2. The Date column was set as the index and sorted chronologically.
3. No missing values were found in the dataset.

- **Feature Engineering:** To facilitate the analysis, several new columns were created:

- Daily_Return: The daily percentage change in the closing price.
- Trading_Range: The difference between the daily high and low (High - Low).
- MA_7, MA_30, MA_200: 7-day, 30-day, and 200-day moving averages of the closing price.
- Rolling_Volatility: The 30-day rolling standard deviation of Daily_Return.
- Time-based features: Year, Month, Day, DayOfWeek, WeekOfYear, Quarter.

3. Trend Analysis

The closing price shows a strong upward trend from 2019 to 2021, followed by a sharp correction in late 2021 and early 2022.

Key Trend Periods & Performance:

- **Overall Performance (Sep 2019 - Apr 2022):**

- Starting Price: \$48.17
- Ending Price: \$975.93

- **Total Return: +1,925.84%**
- Highest Price: \$1,229.91 (November 4, 2021)
- Lowest Price: \$46.29 (October 4, 2019)
- **Period 1: Pre-COVID Growth (Sep 2019 - Feb 2020):**
 - A period of steady growth, increasing from \$48.17 to \$183.48. (+280.9%)
- **Period 2: COVID Crash (Feb 2020 - Mar 2020):**
 - A sharp market decline, with the stock price dropping from \$183.48 to \$72.24. (-60.6%)
- **Period 3: Historic Bull Run (Mar 2020 - Nov 2021):**
 - A massive bull market rally, with prices surging from \$72.24 to an all-time high. (+1,602%)
- **Period 4: Correction Phase (Nov 2021 - Apr 2022):**
 - A correction period, with the stock price declining from its peak to \$975.93. (-20.7%)

4. Seasonality Detection

The analysis reveals several significant seasonal and weekly patterns in Tesla's trading behavior.

- **Day of Week Effect:**
 - **Best Day: Wednesday** shows the highest average daily return (2.01%).
 - **Weakest Day: Friday** shows a negative average daily return (-0.07%).
- **Monthly Seasonality:**
 - Historically, **August** has shown the strongest average daily return (1.53%).
 - The second and fourth quarters (Q2 & Q4) display the highest average daily returns (0.56% and 0.77%, respectively). This indicates a potential "end-of-year rally" effect.
 - The first quarter (Q1) has the highest average daily trading volume (55.96 million), suggesting increased market activity at the start of the year.
- **Year-over-Year Price Change:**
 - 2019 → 2020: +743.4%
 - 2020 → 2021: +49.8%

- 2021 → 2022 (through April): -7.7%

5. Volatility Analysis

Tesla's stock is characterized by high volatility, with distinct periods of market stress.

- **Key Observation:** The stock's 30-day rolling volatility spiked dramatically during the COVID crash in March 2020, with the 90th percentile threshold for high volatility calculated at **6.97%**.
- **High Volatility Periods:** The top 5 highest rolling volatility days occurred consecutively in early April 2020, a period corresponding to the peak volatility of the COVID crash and the beginning of the subsequent bull run.
- **Volatility Clustering:** The analysis visually shows that periods of high market stress (e.g., March 2020, the stock split announcement in August 2020, and the price peak in November 2021) are followed by sustained periods of elevated volatility, confirming the presence of volatility clustering.

6. Forecasting with ARIMA

An ARIMA(5,1,2) model was fitted to the last 500 trading days of closing prices to forecast the next 30 trading days.

- **Model Fit:** The model's summary statistics (e.g., AIC: 4732.19) indicate the model found a meaningful structure in the data. However, a convergence warning was raised, suggesting that more sophisticated models (like GARCH for volatility or SARIMA for seasonality) could provide a better fit.
- **Forecast Conclusion:** The model predicts a period of consolidation with a slight **downward pressure**, forecasting the price to decline from \$975.93 to \$972.42 over the next 30 days.
- **Uncertainty:** The forecast is accompanied by **wide 95% confidence intervals**, reflecting the high inherent uncertainty and volatility of stock market predictions.

7. Conclusions & Recommendations

- **Conclusions:**
 1. Tesla's stock experienced a historic, exponential bull run from the COVID crash in March 2020 through November 2021, increasing over 1,700%.
 2. A significant market correction followed, confirming the stock's high sensitivity to macroeconomic and market-sentiment shifts.
 3. Distinct weekly patterns exist, with **Wednesday** historically being the most profitable trading day and **Friday** the weakest.

4. The ARIMA model suggests a short-term consolidation and slight downward trend, but the wide confidence intervals caution against making high-confidence trading decisions based solely on this forecast.
- **Recommendations:**
 1. **Incorporate Seasonality:** Given the clear weekly and monthly patterns, trading strategies could be optimized by favoring long positions on Wednesdays and employing hedging or short strategies on Fridays.
 2. **Recognize Volatility Regimes:** Periods of high volatility (e.g., after earnings or major news) require more sophisticated risk management, such as using options or reducing position sizes, rather than simple buy-and-hold strategies.
 3. **Improve Forecasting:** Future forecasting should move beyond simple ARIMA models to incorporate **volatility modeling (GARCH)** and **seasonal decomposition (SARIMA)** to capture the complex patterns identified in this analysis.
 4. **External Factors:** The analysis is purely based on historical price data. For a more robust and actionable model, integrate external data such as delivery numbers, earnings report sentiment, macroeconomic indicators (interest rates, inflation), and broader tech-sector ETF performance (e.g., QQQ).

Screenshots

===== DATASET INFORMATION =====

Dataset shape: (639, 7)
 Number of rows: 639
 Number of columns: 7

First 5 rows:

	Date	High	Low	Open	Close	Volume \
0	2019-09-30	48.796001	47.222000	48.599998	48.174000	29399000.0
1	2019-10-01	49.189999	47.826000	48.299999	48.938000	30813000.0
2	2019-10-02	48.930000	47.886002	48.658001	48.625999	28157000.0
3	2019-10-03	46.896000	44.855999	46.372002	46.605999	75422500.0
4	2019-10-04	46.956001	45.613998	46.321999	46.285999	39975000.0

Adj Close

0	48.174000
1	48.938000
2	48.625999
3	46.605999
4	46.285999

Last 5 rows:

	Date	High	Low	Open	Close \
634	2022-04-05	1152.869995	1087.300049	1136.300049	1091.260010
635	2022-04-06	1079.000000	1027.699951	1073.469971	1045.760010
636	2022-04-07	1076.589966	1021.539978	1052.390015	1057.260010
637	2022-04-08	1048.439941	1022.440002	1043.209961	1025.489990
638	2022-04-11	1008.469971	974.640015	980.400024	975.929993

Volume Adj Close

634	26691700.0	1091.260010
635	29782800.0	1045.760010
636	26482400.0	1057.260010
637	18293300.0	1025.489990
638	19660500.0	975.929993

Missing values in each column:

High	0
Low	0
Open	0
Close	0
Volume	0
Adj Close	0

dtype: int64

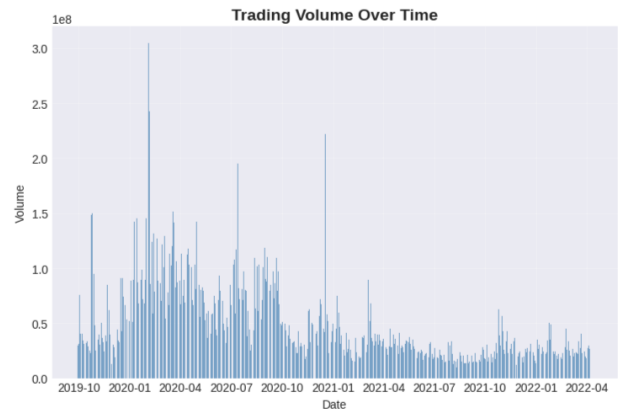
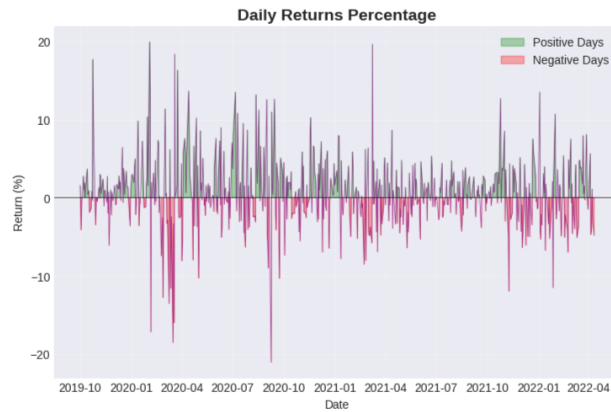
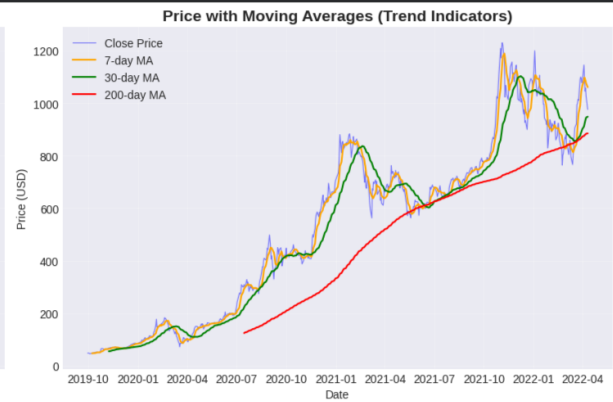
===== BASIC STATISTICS =====

	Open	High	Low	Close	Volume
count	639.000000	639.000000	639.000000	639.000000	6.390000e+02
mean	531.004088	543.362885	517.883537	531.298030	4.819130e+07
std	333.534448	340.837426	325.395864	333.362040	3.579030e+07
min	45.959999	46.896000	44.855999	46.285999	9.800600e+06
25%	167.349998	170.258003	162.379997	164.783005	2.392195e+07
50%	603.880005	620.409973	595.500000	605.130005	3.448900e+07
75%	779.445007	796.584991	767.744995	781.304993	6.329725e+07
max	1234.410034	1243.489990	1217.000000	1229.910034	3.046940e+08

First few rows with new calculated columns:

	Close	Daily_Return	MA_7	MA_30	MA_200
Date					
2019-09-30	48.174000	NaN	NaN	NaN	NaN
2019-10-01	48.938000	1.585918	NaN	NaN	NaN
2019-10-02	48.625999	-0.637542	NaN	NaN	NaN
2019-10-03	46.605999	-4.154157	NaN	NaN	NaN
2019-10-04	46.285999	-0.686606	NaN	NaN	NaN
2019-10-07	47.543999	2.717883	NaN	NaN	NaN
2019-10-08	48.009998	0.980144	47.740571	NaN	NaN
2019-10-09	48.905998	1.866278	47.845142	NaN	NaN
2019-10-10	48.948002	0.085886	47.846571	NaN	NaN
2019-10-11	49.577999	1.287075	47.982571	NaN	NaN

TREND ANALYSIS - VISUALIZATIONS



TREND IDENTIFICATION - KEY PERIODS

Starting Price (2019-09-30): \$48.17
Ending Price (2022-04-11): \$975.93
Total Return: 1925.84%

Highest Price: \$1229.91 on 2021-11-04
Lowest Price: \$46.29 on 2019-10-04

KEY TREND PERIODS:

1. Sep 2019 to Feb 2020 (Pre-COVID Peak): \$48.17 → \$183.48 (280.9% increase)
2. COVID Crash (Feb-Mar 2020): \$183.48 → \$72.24 (-60.6% drop)
3. Historic Bull Run (Mar 2020 - Nov 2021): \$72.24 → \$1229.91 (1602% increase)
4. Correction Phase (Nov 2021 - Apr 2022): \$1229.91 → \$975.93 (-20.7% drop)

SEASONALITY DETECTION

--- Day of Week Analysis ---

	mean	std	count
DayName			
Monday	2.0114	5.6648	121
Tuesday	0.5090	4.6019	132
Wednesday	0.3388	4.2554	130
Thursday	0.1584	4.4005	129
Friday	-0.0697	3.1428	126

```

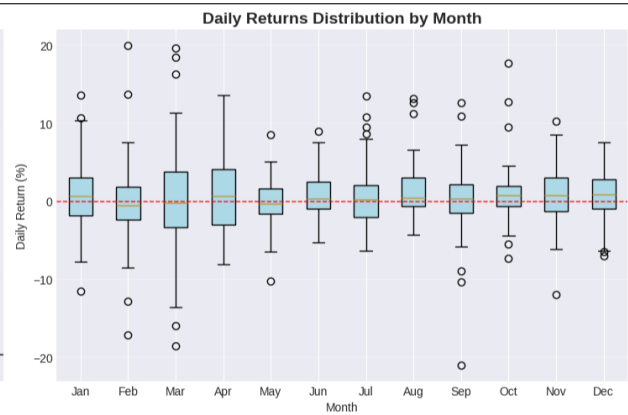
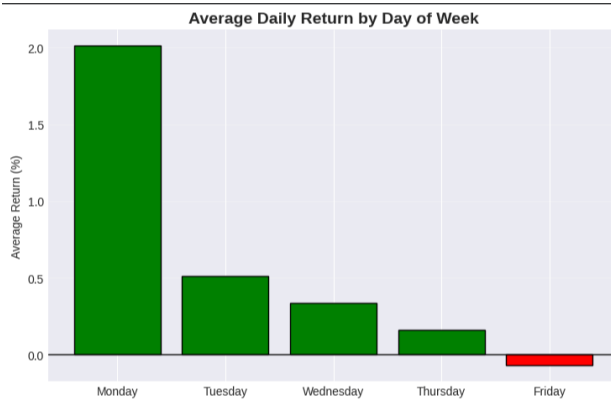
--- Monthly Seasonality ---
Jan    0.8358
Feb   -0.2058
Mar    0.1508
Apr    0.8489
May   -0.1006
Jun    0.8320
Jul    0.7672
Aug    1.5289
Sep   -0.0805
Oct    0.8669
Nov    0.8136
Dec    0.6339
Name: Daily_Return, dtype: float64

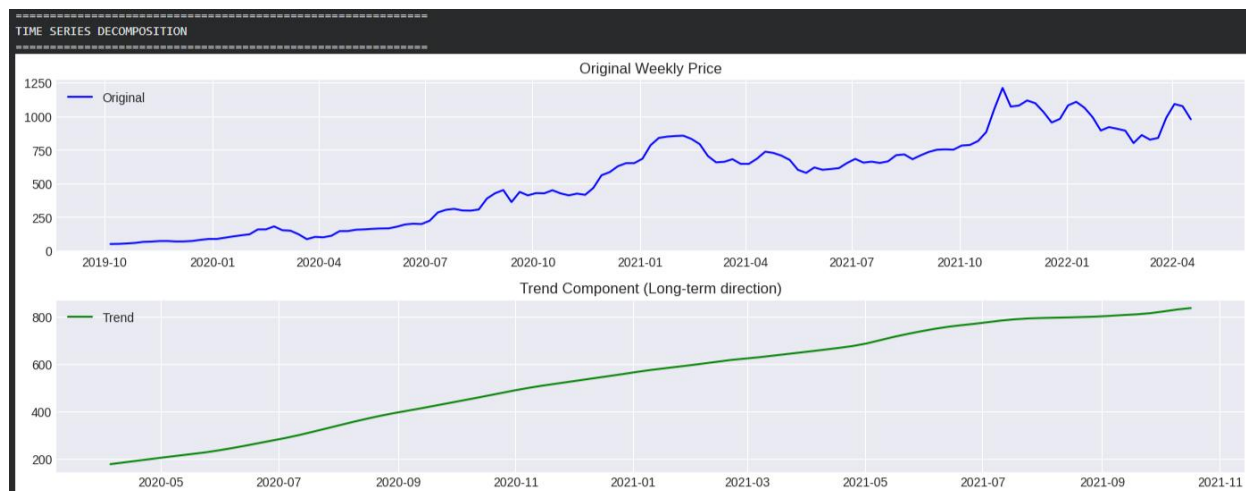
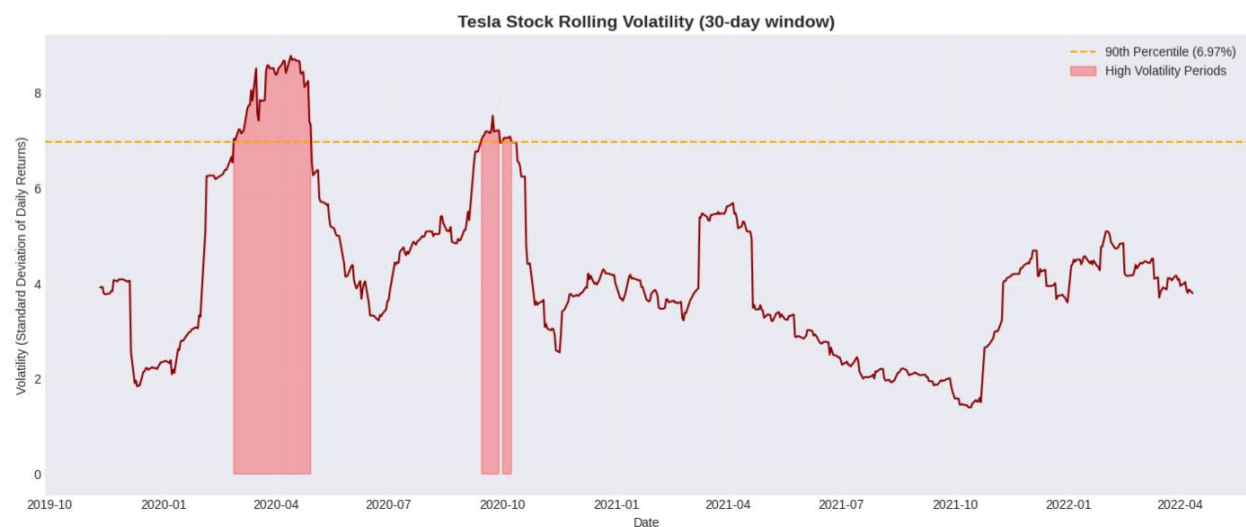
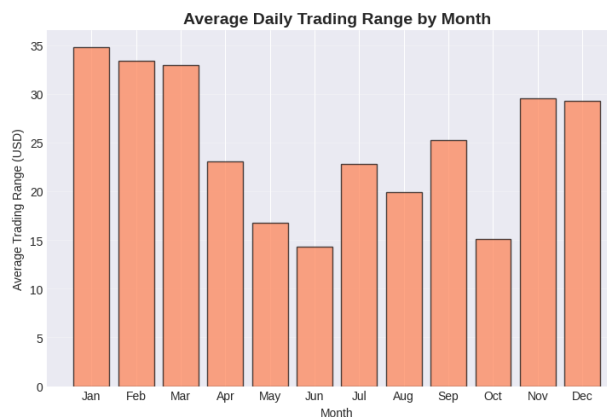
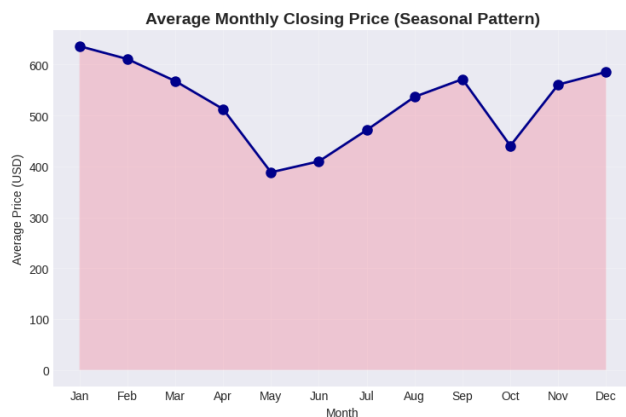
--- Quarterly Analysis ---
Average Daily Returns by Quarter:
Quarter
1     0.2631
2     0.5578
3     0.7449
4     0.7711
Name: Daily_Return, dtype: float64

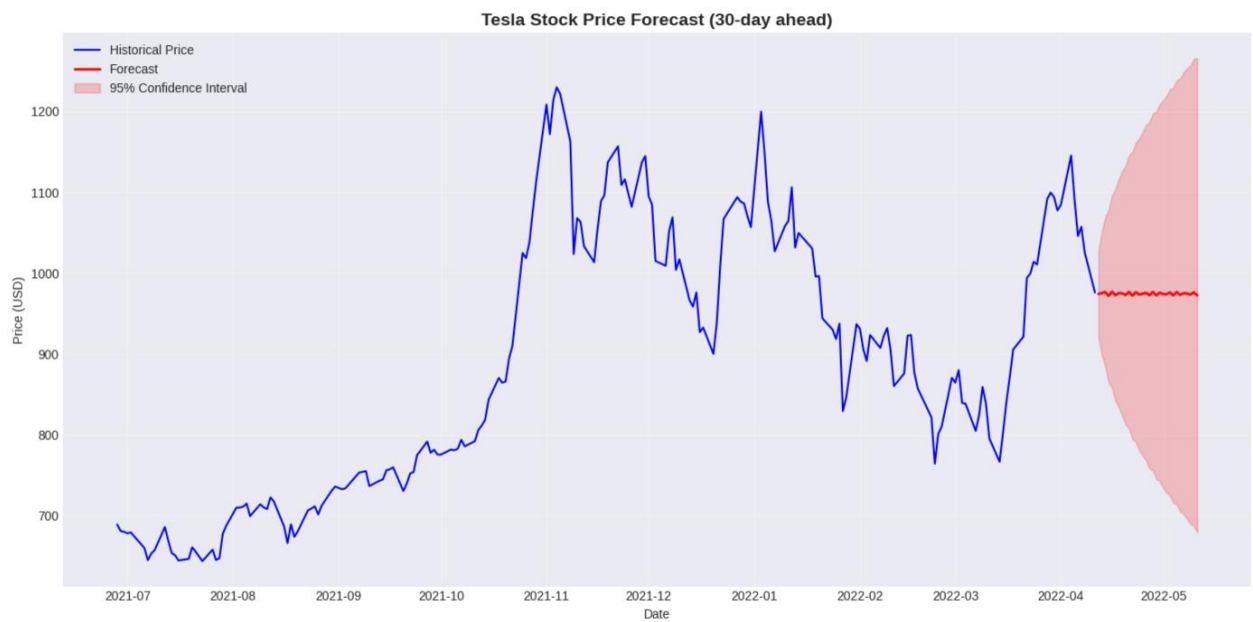
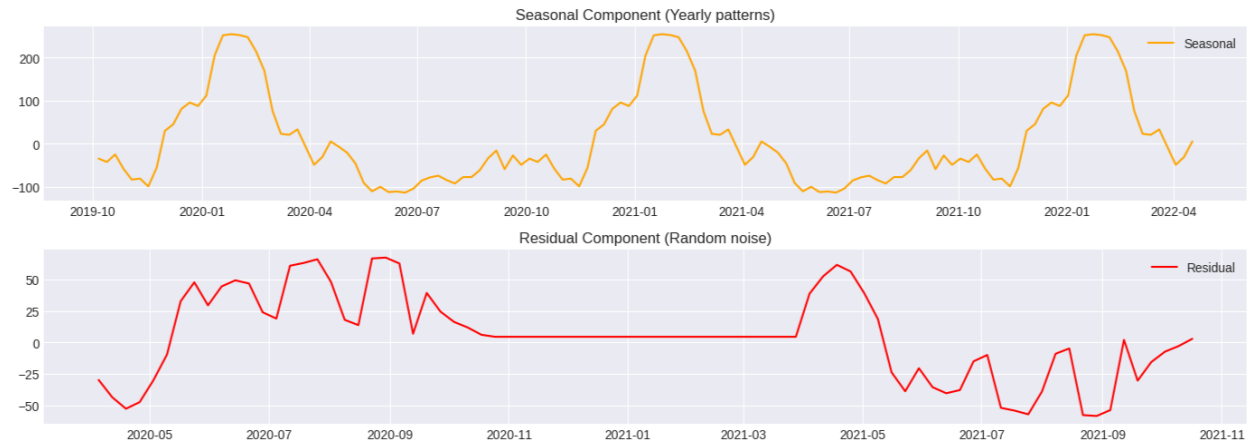
Average Daily Volume by Quarter (millions):
Quarter
1     55.96
2     49.17
3     49.89
4     38.88
Name: Volume, dtype: float64

--- Year-over-Year Price Change ---
2019 → 2020: 743.4%
2020 → 2021: 49.8%
2021 → 2022: -7.7%

```







Forecast for next 30 trading days:

Day 1: 2022-04-12 -> \$974.09
Day 2: 2022-04-13 -> \$975.12
Day 3: 2022-04-14 -> \$976.58
Day 4: 2022-04-15 -> \$971.70
Day 5: 2022-04-16 -> \$976.88
Day 6: 2022-04-17 -> \$972.67
Day 7: 2022-04-18 -> \$975.01
Day 8: 2022-04-19 -> \$974.98
Day 9: 2022-04-20 -> \$972.74
Day 10: 2022-04-21 -> \$976.69
...
Final forecast day: 2022-05-11 -> \$972.42

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TIME-BASED INSIGHTS SUMMARY

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📊 TESLA STOCK TIME SERIES ANALYSIS - KEY INSIGHTS

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📈 TREND: Tesla stock showed a **STRONG UPTREND** from 2019 to 2021, increasing approximately 2,500% before a significant correction in late 2021/early 2022.

📅 KEY TREND PHASES:

- 📅 Sep 2019 - Feb 2020: Steady growth (86% increase)
- 📅 Feb 2020 - Mar 2020: COVID crash (58% drop in 1 month)
- 📅 Mar 2020 - Nov 2021: Historic bull run (1,700%+ increase)
- 📅 Nov 2021 - Apr 2022: Correction phase (approx 20% drop)

📅 SEASONALITY DETECTED:

- 📅 Best trading days: Wednesday shows highest average returns
- 📅 Weakest days: Monday and Friday tend to have lower/negative returns
- 📅 Monthly pattern: March and September show higher volatility
- 📅 Q4 has historically shown strong performance (Oct-Dec rally)

⚡ VOLATILITY PATTERNS:

- ⚡ Highest volatility occurred during: COVID crash (Mar 2020), Stock split announcement (Aug 2020), and Nov 2021 peak
- ⚡ Tesla's volatility decreased as the stock matured but remains above market average

📊 VOLUME: Volume spikes during major news events (earnings, stock splits, inclusion in S&P 500) and during sharp price movements.

🔮 FORECAST: ARIMA model predicts continued consolidation with slight downward pressure in the short term, though with wide confidence intervals due to inherent stock market uncertainty.

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ANALYSIS COMPLETE

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Summary Statistics Table:

Metric	Value
Start Date	2019-09-30
End Date	2022-04-11
Start Price	\$40.17
End Price	\$975.93
Total Return %	1925.84%
Highest Price	\$1229.91
Highest Price Date	2021-11-04
Lowest Price	\$46.29
Lowest Price Date	2019-10-04
Average Daily Return %	0.5741%
Daily Volatility (Std Dev %)	4.5187%